

STUDY GUIDE ROB 2040 FINAL

Printouts of the following Studio 5000 labs will be helpful:

SS1

SS2

TIMED WASHER STATION RETENTIVE TIMER

Printouts of the following SIEMENS labs will be helpful:

STARTUP TIMER

CASCADING CONVEYOR TIMER

SPOOL

All Studio 5000 and SIEMENS Programs

Type the Tag name correctly.

Equivalent terms

- Set = voltage = Logic 1
- Reset = no voltage = Logic 0

Studio 5000 Programs

Notes on Tags:

- Notice which tags are Controller Tags and which are Program Parameter and Local Tags and assign correctly.
- Change to the correct Data Types for Boolean Tags.
- Change to the correct Data Types for Timers and Counters.

SOLENOIDS - Studio 5000

(Not based on a lab but equal in context to the labs)

The program will use a Stop/Start circuit to turn on the first solenoid immediately and a second solenoid after a time delay.

Press and release the **START** pushbutton to:

- The timer's Enable bit is used in the holding circuit.
- The timer's Enable bit is used to energize the first solenoid.
- The timer's Done bit delays energizing the second solenoid.

Note: The .EN and .DN are automatically created when the START_TIMER Data Type Timer tag is created.

Two indicators will inform if the if the system has been started (powered ON) or stopped (powered OFF).



Hint: similar to the ON/OFF indicator is SS1.

What timer status bit would be most appropriate for controlling the ON/OFF indicators?

Pressing of the **STOP** pushbutton turns off both solenoids.

WASHER - Studio 5000

The problem is based on the lab, WASHER STATION WITH RETENTIVE TIMER.

Logic is added to the Clamp Solenoids output rung to *hold* the clamps engaged, if the Start pushbutton is released before the end of the timing cycle.

That is, the releasing of the Start pushbutton or if the washing cycle is not complete, the solenoids remain *held* On.

Hint: Similar to SS2 (Jog/Run Lab), the holding circuit is disabled (in this program, when the washing cycle is complete).

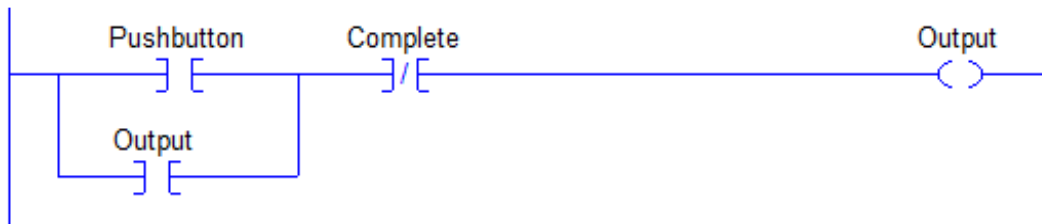
Holding Circuit Examples

Example 1:

In this example rung with the Complete is in series with the holding circuit and the Output will turn ON when the Pushbutton is pressed (True) and the Complete is True.

When the pushbutton is released and while Complete is True, the output remains ON.

When the Complete is False, the Output turns OFF even if the Pushbutton is pressed.



continue

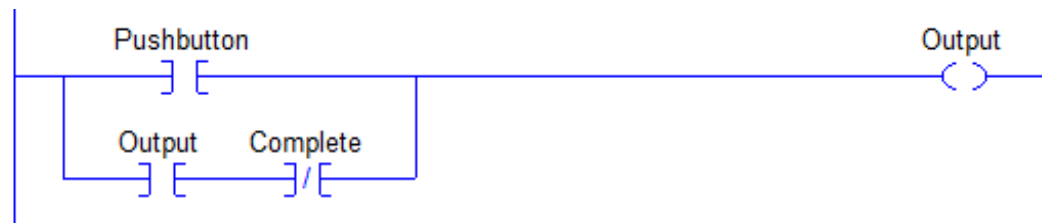


Example 2:

In this second example, Complete is in the holding circuit. The pressing of the Pushbutton turns On the Output.

The Output will be ON if the Pushbutton is released and will remain on if Complete is True and the Output Tag in the holding is True.

The difference is when Complete is False the holding circuit is disabled and the Output would remain ON only if the Pushbutton pressed.



A Counter is added that will count the part when:

- the pushbutton is released,
- the clamps are opened,
- and the washing timing cycle is complete.

The only function of this counter is to count the parts.

A reset pushbutton returns the counter to zero.

CLAMPS - Studio 5000

(Not based on a lab but equal in context to the labs)

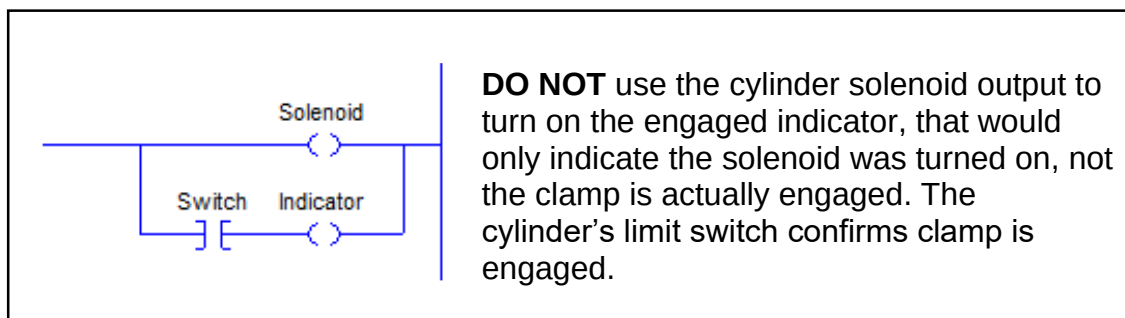
This problem has four (4) cylinders that clamp a part. Each clamps have an Engaged and Disengaged Limit Switch.

Rung 0 to 3

The Clamp Solenoids are energized to hold the part for the machining process.

- When the Part In Position detect (voltage signal) the part **and** while the machine signals the Machining Process (is ***not***) Complete (no voltage signal) the clamp solenoids close on the part.

- Each cylinder will turn on an Indicator when its clamp is engaged. There is an input which directly provides a signal confirming the state, use the Clamp Engaged Limit Switch input.
- All Solenoids and All Indicators are controlled by the same Part In Position and Machining Process (is **not**) Complete and those Outputs can be in a Parallel structure.
- The Indicators will have the Switch from the clamp cylinder in series with the branch's Output. **See illustration below.**



The program has handshaking with a machine to begin the machining process when the clamps are holding the part and when the machining process is complete to release the part.

Rung 4

- If the part is in position **and all** the Clamps Are Engaged, the PLC signals the machine to Start the Machining Process.

Rung 5

- When the machine signals the Machining Process is Complete **and all** the Clamps Are Disengaged, the PLC signals the removal of the part.

continue

DELAYED - Siemens

This is a combination of the horn **STARTUP TIMER** and the **CONVEYOR TIMER** labs.

The change from the TIMED CONVEYOR lab is the first output is now delayed to energize, turn on, after the horn has sounded.

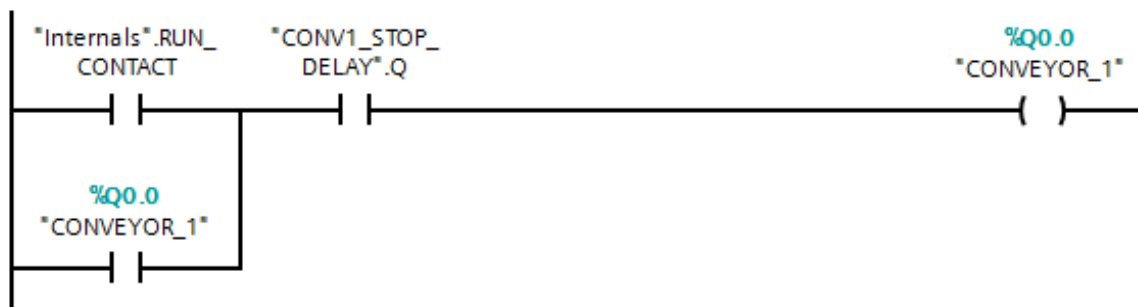
In the FINAL, similar to the STARTUP TIMER lab...

The Stop/Start pushbuttons starts the timer elapsing, the timers Enable bit is used in the holding circuit.

The Horn output is controlled by the timer's status bit(s).

In the FINAL, the starting of the first Output (Conveyor 1 in TIMED CONVEYOR) and delay timer for the second Output (timer for Conveyor 2 in TIMED CONVEYOR) changes due to the Start/Stop for the Horn.

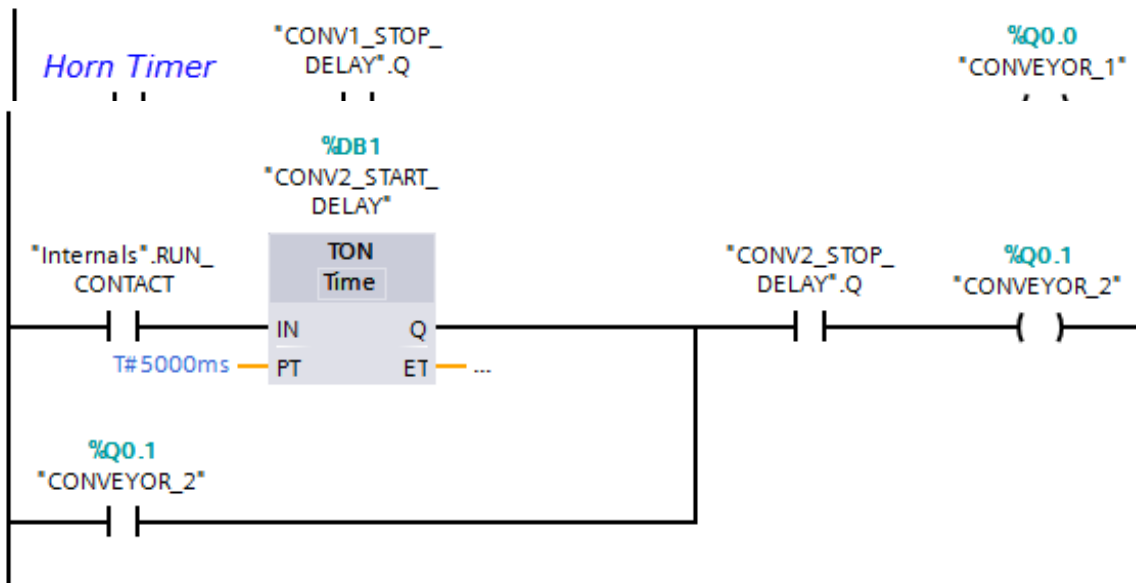
In the LAB, CONVEYOR TIMER the Motor for Conveyor 1 is started by the internal output from the Stop/Start rung.



In the FINAL, the Stop/Start must first sound the horn. When the horn sounding is complete, the horn's timer status bit starts the first conveyor.

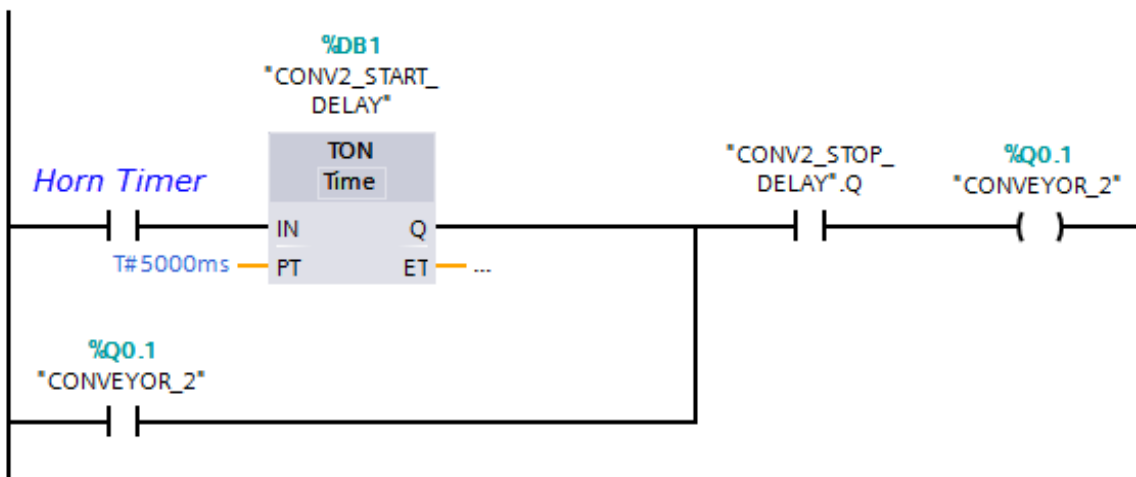
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In the CONVEYOR TIMER, the Run Contact from the Start/Stop pushbutton begins the elapsing of the timer for the Conveyor 2 to delay energizing.

In the FINAL, the Run Contact is substituted with the horn's timer status bit to start elapsing the timer the second conveyor.



In the FINAL, as with CONVEYOR TIMER, when the STOP pushbutton is pressed and released, the first conveyor output is delayed turning Off. The second conveyor output is delayed Off after the first output is Off.

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SPOOLING - Siemens

The SPOOLING final is modified from the **SPOOL COUNTER** lab.

Modifications to the program:

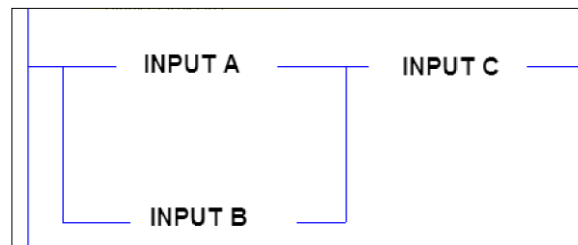
The Spooling Motor can be energized by the Start of the Stop/Start pushbuttons if the Kit Counter **and** Grand Total Counter have **not** reached their count preset.

The Kit Counter will be reset either **automatically** at the end of the Kit Counter count **or** by a Reset pushbutton (switch) **or** by an HMI pushbutton.

The Grand Total Counter is reset by a Reset pushbutton (switch) **or** by an HMI pushbutton.

For all of the resets above the Counter will only be reset if the Spool Motor is **not** energized.

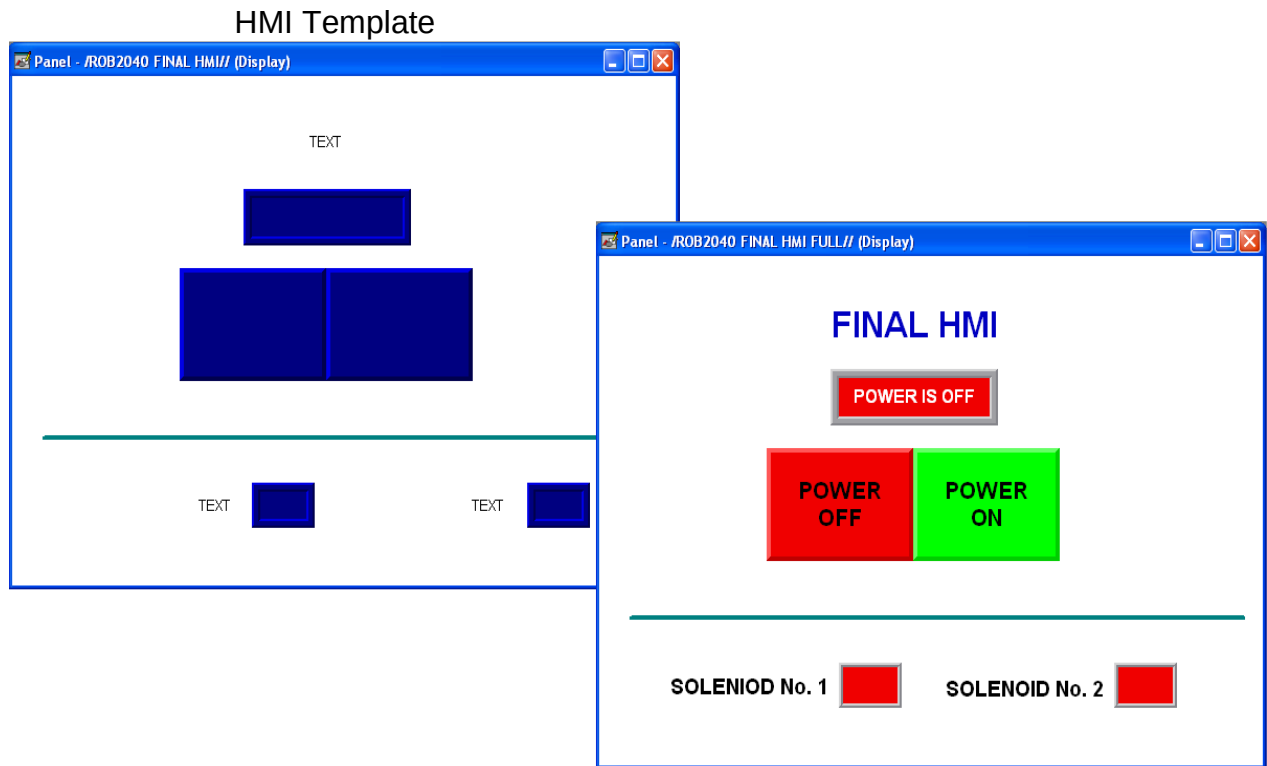
The circuit description above is a of an OR - AND logic and the illustration below showing one of the reset branches.



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HMI

The HMI will use the Final **Solenoid** problem.



Procedures and objects required for HMI.

- Creating the shortcut.
- Assigning PLC program for Offline Development.
- Text objects properties.
- Momentary pushbuttons properties.
- Multistate indicators properties.
- Assigning Tags to the pushbuttons and indicators.
- Assigning PLC for Runtime Design Communications.